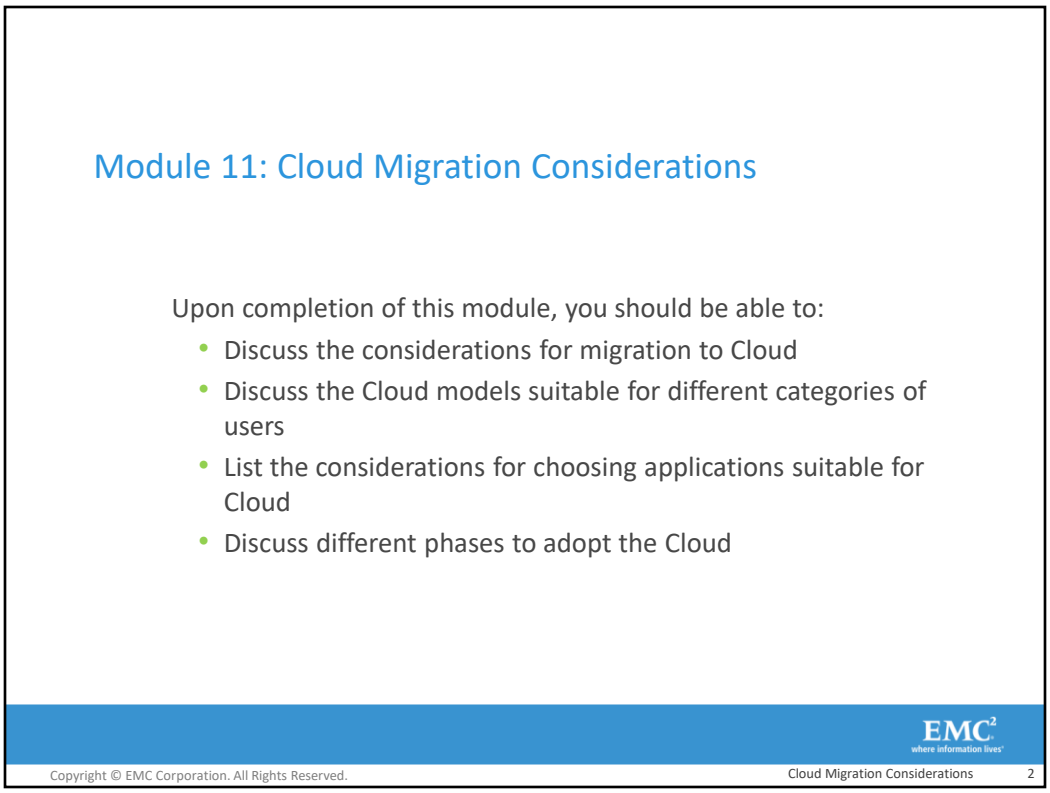
Module – 11

Cloud Migration Considerations

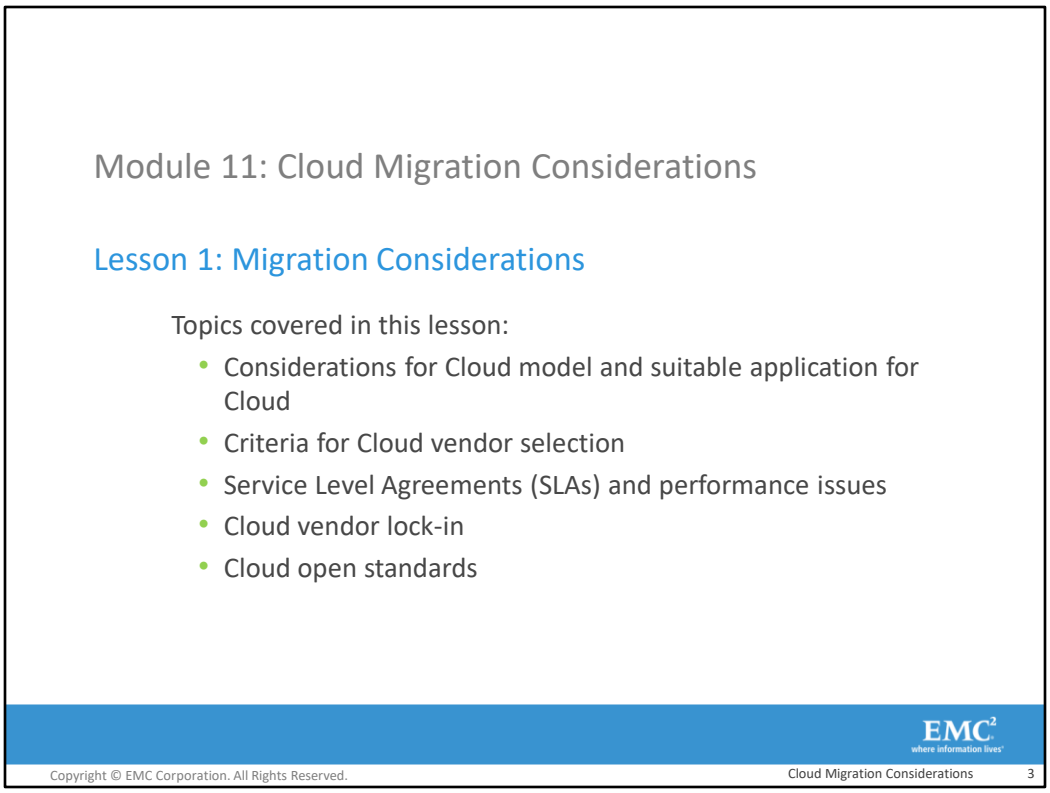
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This module focuses on considerations for migration to the Cloud. It details 'Cloud model' suitable for different categories of users. Further, it covers considerations for choosing candidate application and various other considerations for migration to Cloud. It also covers various phases to adopt the Cloud.

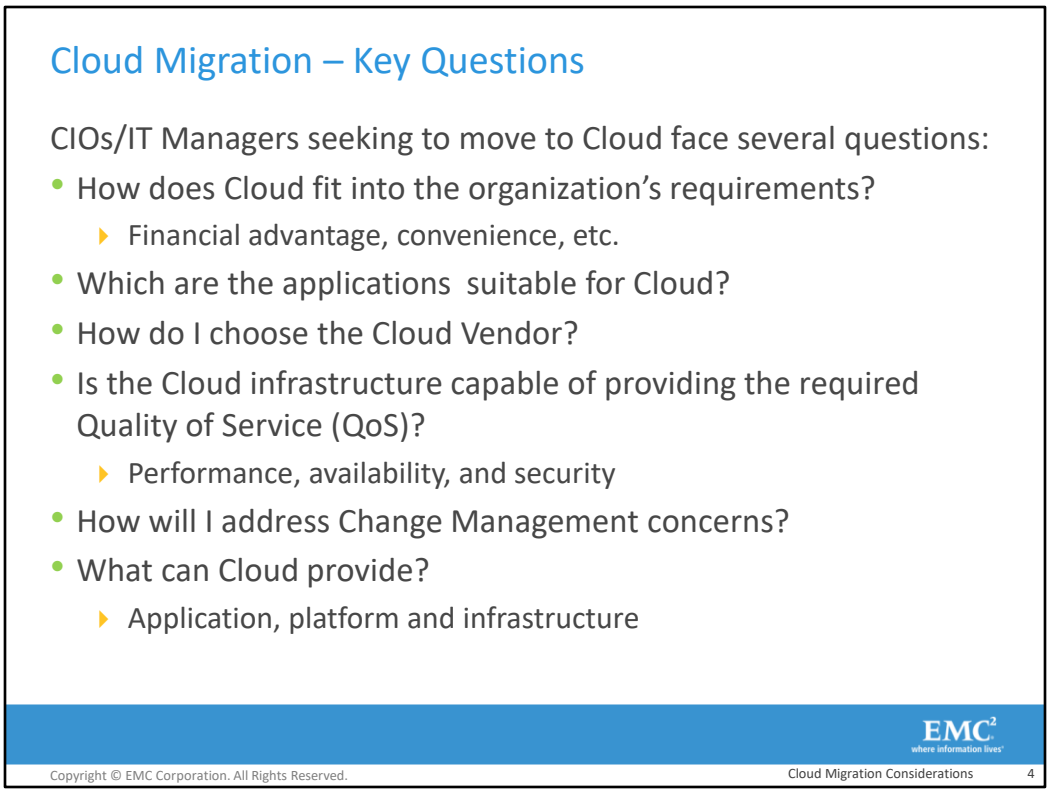
Note: All the discussions in this module are primarily focused on Public Cloud and external private Cloud, referred here as "Cloud". However, the discussed approach may be considered while migrating to Internal private Clouds.

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This lesson covers the key points to consider while migrating an application to the Cloud. The key points to consider are: How Cloud fits in an organizations environment, Cloud model, candidate applications, criteria for Cloud vendor selection, Service level agreements, etc.

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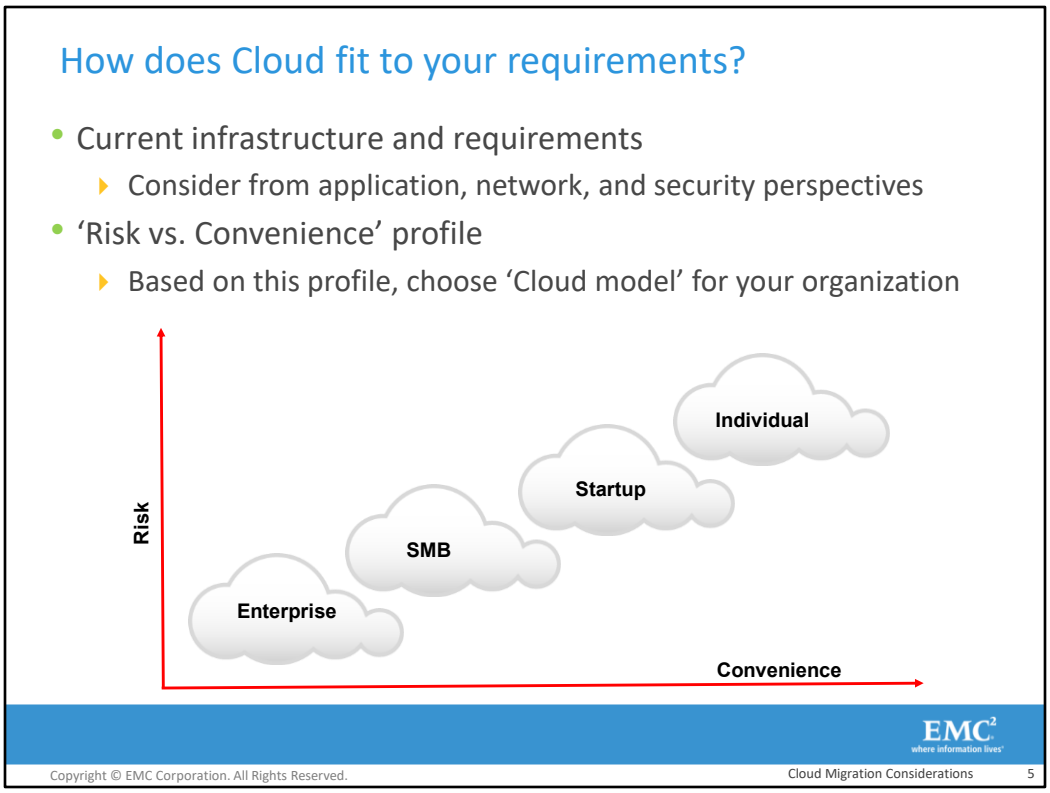


Organizations are not only looking to get a financial advantage with their first move into Cloud, but are also making a significant learning experience to expand their Cloud perspective. Businesses, determining to make their first move into the Cloud, always face a question "How does Cloud fit to the organization's environment?" This is so because there is a risk of introducing an evolving Cloud into an established system.

Most companies are not ready to abandon their existing IT investments to move all of their business processes fully to the Cloud at once. Instead, it is more likely to be a gradual shift in the business processes to the Cloud over time. The reason behind this cautious approach is that the Cloud providers are not assuring the same levels of security, controls, and performance that organizations have on premises. Lack of regulatory compliance and policies for both providers and consumers further slow down the adoption of Cloud.

It is important to understand the various Cloud migration considerations before migrating to the Cloud.

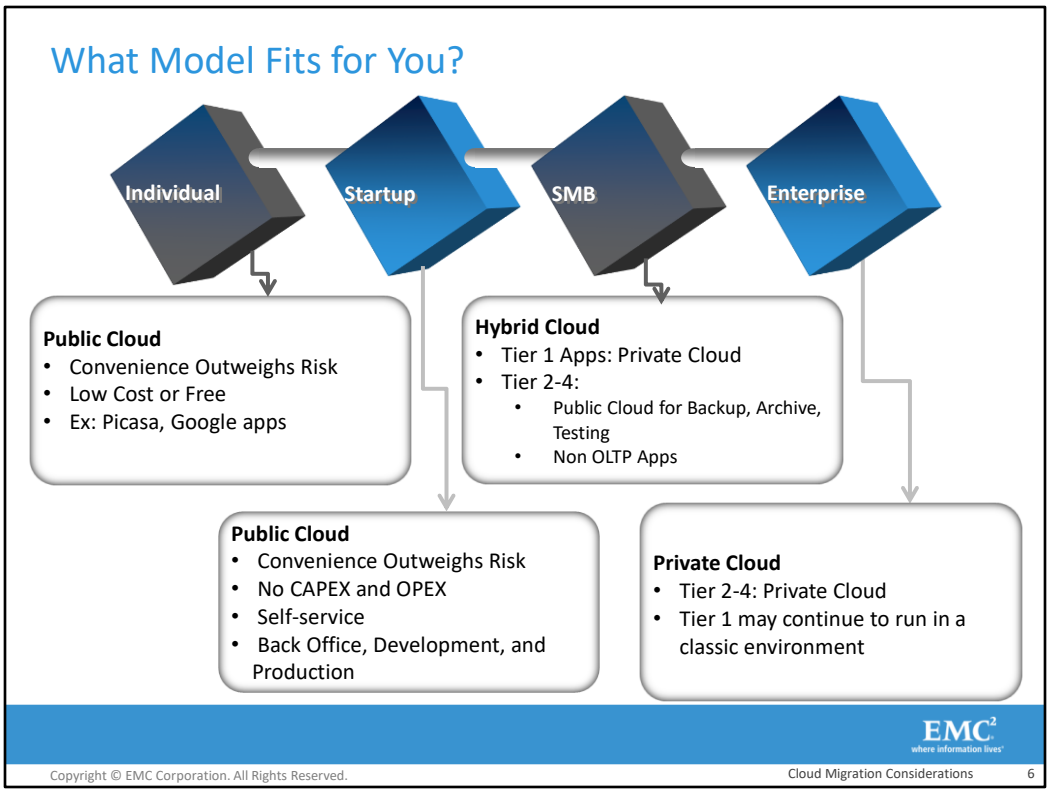
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Studies based on the experience of early Cloud adopters suggest that moving to the Cloud without proper strategy and process does not yield expected benefits. The most important concern that needs to be evaluated before making a move to Cloud is 'How does Cloud Computing fit in the context of the organization's overall business strategy?". Sometimes, a Cloud may look attractive from an application perspective. However, in real time, it could be a challenge for network administrators. Further more, there could be security concerns about having data outside the firewall.

Risk versus convenience is a key consideration for deciding Cloud migration strategy. This consideration also forms the basis for choosing right Cloud model. Cloud benefits are well established, but data may reside outside the organization's perimeter causing risk. A balance must be evaluated to determine how much risk an individual or organization may handle for the benefit of convenience. This proportion varies among Cloud consumers, based on which they may be segmented into individual, business startup, small and medium business, and enterprise. Typically, individuals and startup businesses are ready to take high risk to get most of the convenience offered by a Cloud. Compared to that, SMBs and enterprises are more sensitive to risk and are unlikely to move their applications to Cloud.

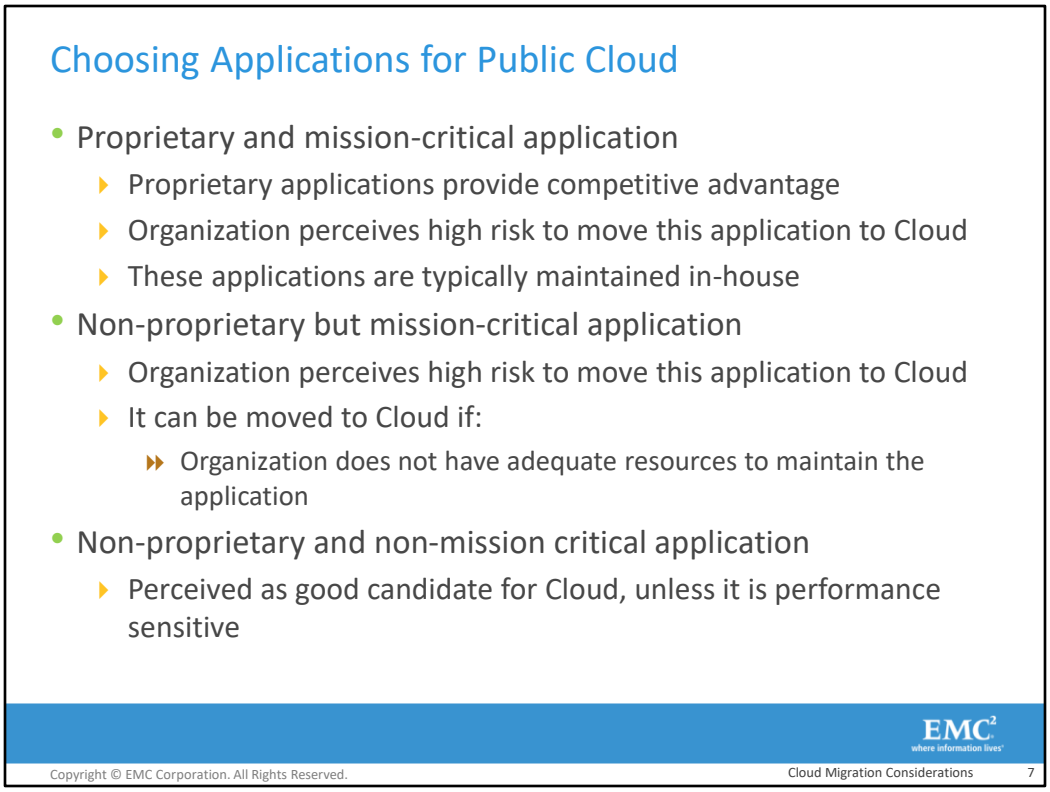
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Let us understand which Cloud model will be most suitable for an organization or an individual:

- Public Cloud is preferred by individuals who want to access Cloud services such as Picasa and Google apps, and are least concerned about the security or availability risks in Cloud for the most part. Here, cost reduction is the primary objective. Public Cloud enables the opportunity to access these applications for free or by paying minimum usage charges.
- People who start up businesses from small office or home typically opt for Public Cloud. A large investment to purchase IT resources is not affordable or may not give the required ROI. Therefore, for obvious reasons, convenience offered by the Cloud outweighs risk.
- Small and medium-sized businesses have a moderate customer base and any anomaly in customer data and service levels may impact their business. Hence, they may not be willing or be able to put Tier 1 applications, such as Online Transaction Processing (OLTP), in the Cloud. A hybrid Cloud model may fit in this case, which includes the organization's internal IT resources (Private Cloud) and external Public Cloud resources. Tier 1 application data should never cross the boundary of Private Cloud. Public Cloud enables cost savings and faster time to market and is typically used for tier 2, tier 3, and tier 4 applications such as backup, archive, and testing.
- Enterprises typically have a strong customer base worldwide. The priority is to maintain critical customer data and service levels, with strict enforcement of security policies. They are highly concerned with the risk and information access control in Cloud. They are financially capable of building a massive Private Cloud. Many enterprises may not even want to move any of their applications to Cloud.

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Not all applications are good candidates for Cloud, although it may depend on the capability of the Cloud infrastructure and the quality of service offered by Cloud providers.

When migrating applications to the Cloud, there are three general considerations that may be used to determine if the application can move to the Cloud. Proprietary and mission-critical applications are core and essential to the business. Often, they are applications that provide competitive advantages and are usually designed, developed, and maintained in-house. Typically, the perceived risk and effort to outsource these systems to the Cloud is high.

Give close consideration to applications that are non-proprietary, but are still mission-critical. Though the effort to migrate these applications to the Cloud may be minimal, the perceived risk to the business may be deemed high. If the organization does not have adequate resources to maintain the application or if the cost to maintain the application is high, then this may outweigh the risks.

The sweet spot for migrating applications to the Cloud is the non-proprietary and non-mission critical applications if they are not performance sensitive. These applications have good compatibility, standardized functionality, and interfaces, making the level of migration effort minimal in comparison to proprietary applications. Since these are non-proprietary and non-mission critical applications, moving to the Cloud poses minimum risk.

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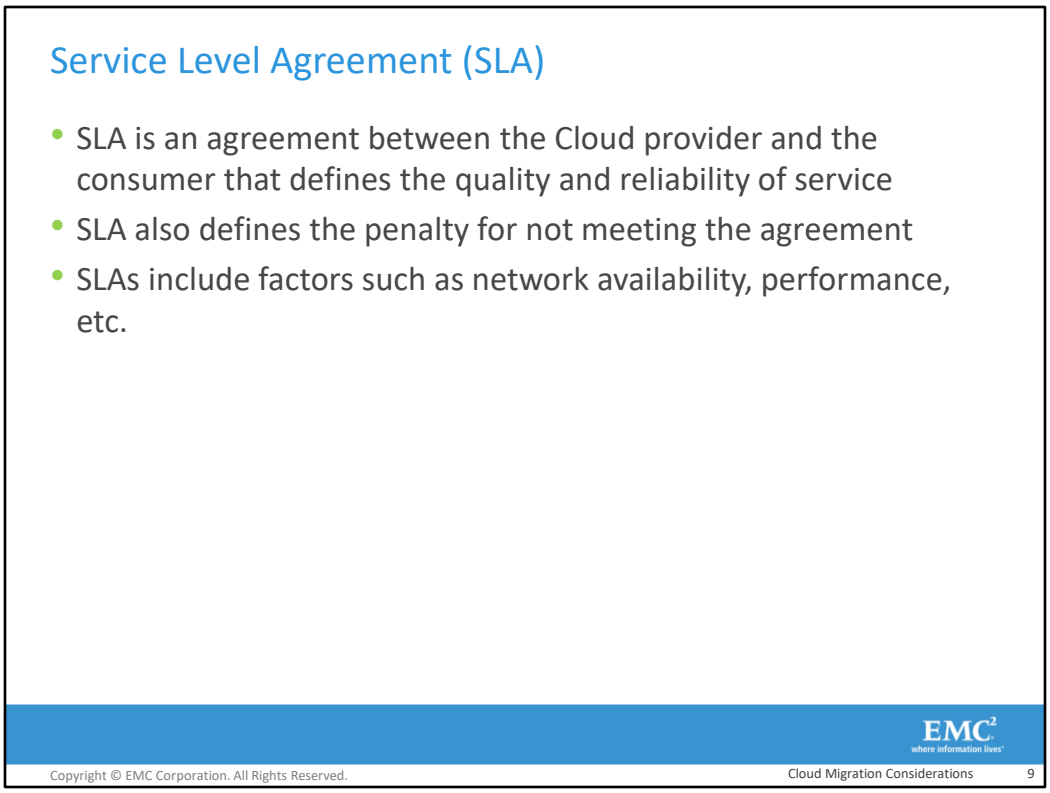
Selecting a Cloud Service Provider

- Some key questions to ask before selecting a Cloud Service provider:
  - How long has the provider been providing the services?
  - How well does the provider meet the organization's current and future requirements?
  - How easy is it to relinquish resources not in use and to reduce cost?
  - What tools does the provider offer (like virtual machine images) that would make it easy to move to another provider when required?
  - How easy is it to add/remove services?
  - Does the provider have good customer service support?
  - What happens when the provider upgrades their software? Is it forced on everyone? Can you upgrade on your own schedule?
  - Does the provider offer required security services?
  - Does the provider meet your legal and privacy requirements?

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Cloud is an emerging technology and many Cloud players are just entering the market. Out of the several Cloud service providers, selecting a provider is a critical task. Some key provider selection criteria are listed on this slide.

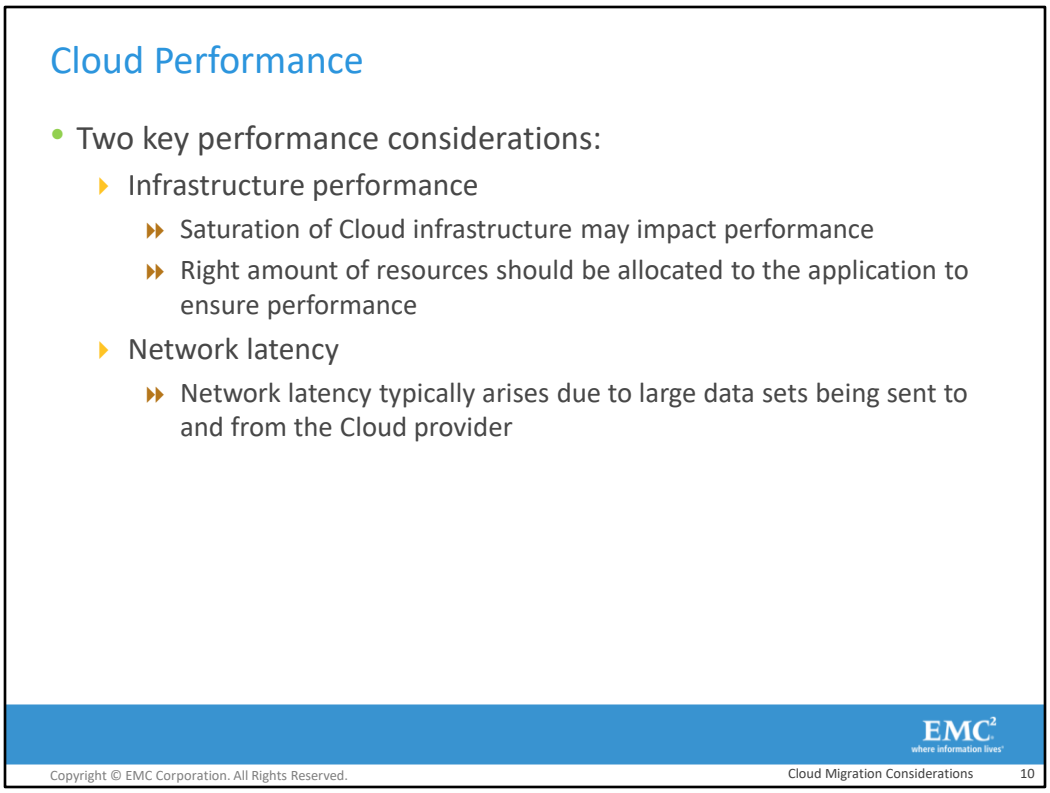
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As consumers move towards Cloud, the quality and reliability of services become important considerations. However, the demands of the consumers vary significantly. It is not possible to fulfill all consumer expectations from the service provider's perspective, and hence, a balance needs to be made via a negotiation process. At the end of the negotiation process, the provider and the consumer commit to an agreement. This agreement is referred to as Service Level Agreement (SLA). This SLA serves as the foundation for the expected level of service between the consumer and the provider. The QoS (Quality of Service) attributes are generally part of an SLA (such as response time and throughput). However, these attributes change constantly, and to enforce the agreement, these parameters need to be closely monitored.

Strong Service Level Agreements (SLAs) from Cloud vendors are a must to ensure QoS. Without these agreements and penalties for failing to meet them, vendors have less incentive to maintain performance at the highest levels. SLAs can include factors such as network availability, performance, etc.

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There are two key factors that impact Cloud performance, infrastructure performance and network latency.

- **Infrastructure performance:** Most Cloud platforms leverage a shared, multitenant, virtual infrastructure. An application may have its own virtual space or virtual machine, but it shares processors and storage space with several other applications on that Cloud infrastructure. It is possible that the Cloud infrastructure may become saturated from time to time, and thus impact performance. There is not much that can be done about this, other than work with the Cloud provider to ensure that the application gets the required performance. Typically, this is taken care in the SLA.
- **Network latency:** Performance related to network latency typically arises due to large data sets being sent to and from the Cloud provider. The larger the dataset, the more likely that the network performance issues come into play.

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Cloud Vendor Lock-in

- Cloud vendor (service provider) may lack open standards or use proprietary software/APIs
- Rigid agreements prevent the consumer from moving without penalties
- Cloud vendors may prevent a consumer from moving one service model to another (i.e. application built on a PaaS moving to an IaaS model)
- Application may require significant rework/redesign before deploying in different Cloud

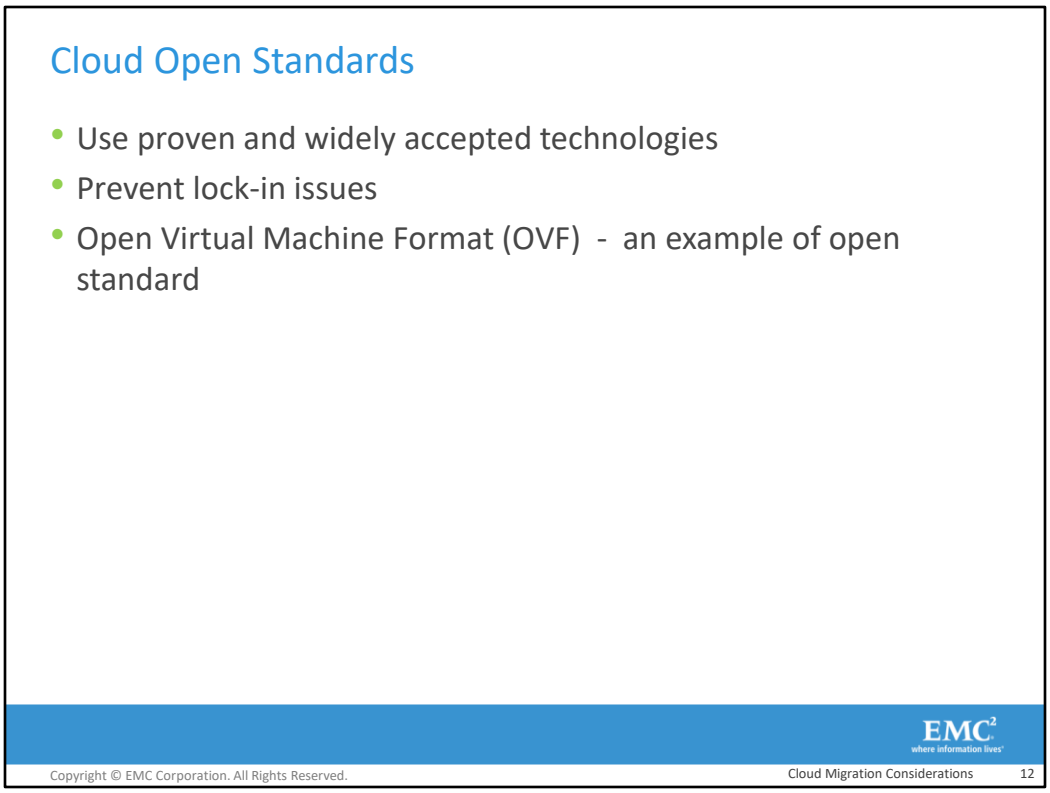
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This slide lists some key reasons that may prevent moving an application from one Cloud to another.

Cloud lock-in refers to a situation where a Cloud consumer is unable to move out of the current Cloud vendor (service provider) due to the complexity/restriction imposed by the current Cloud vendor.

Vendor lock-in is seen as deterrent to moving services to the Cloud. This concern is magnified when looking at a federated Cloud. Lock-in may prevent an organization from moving their application from one Cloud to another to take advantage of geography and for potential performance improvement. It makes a customer dependent on a vendor because switching to another vendor may come with costs.

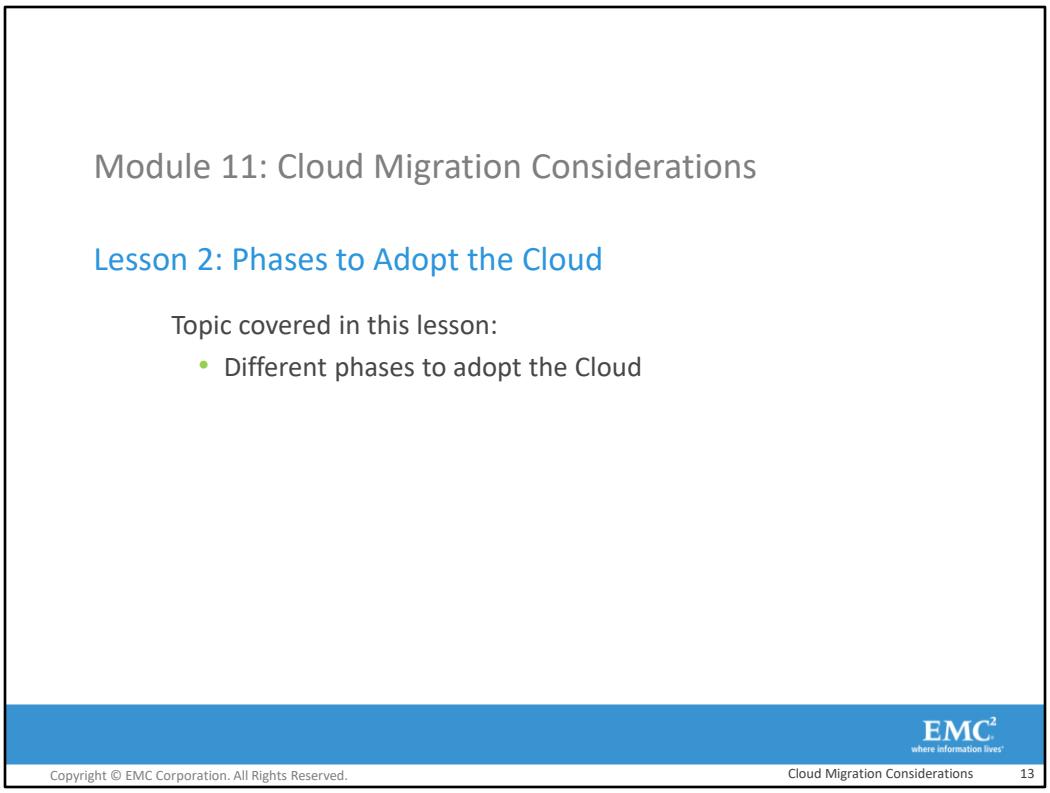
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Open standards are the building block for multi-vendor, federated Clouds and can make vendor lock-in avoidable. Widely accepted standards provide interoperability and portability. Without open standards, it becomes difficult to connect public, private, and hybrid Clouds.

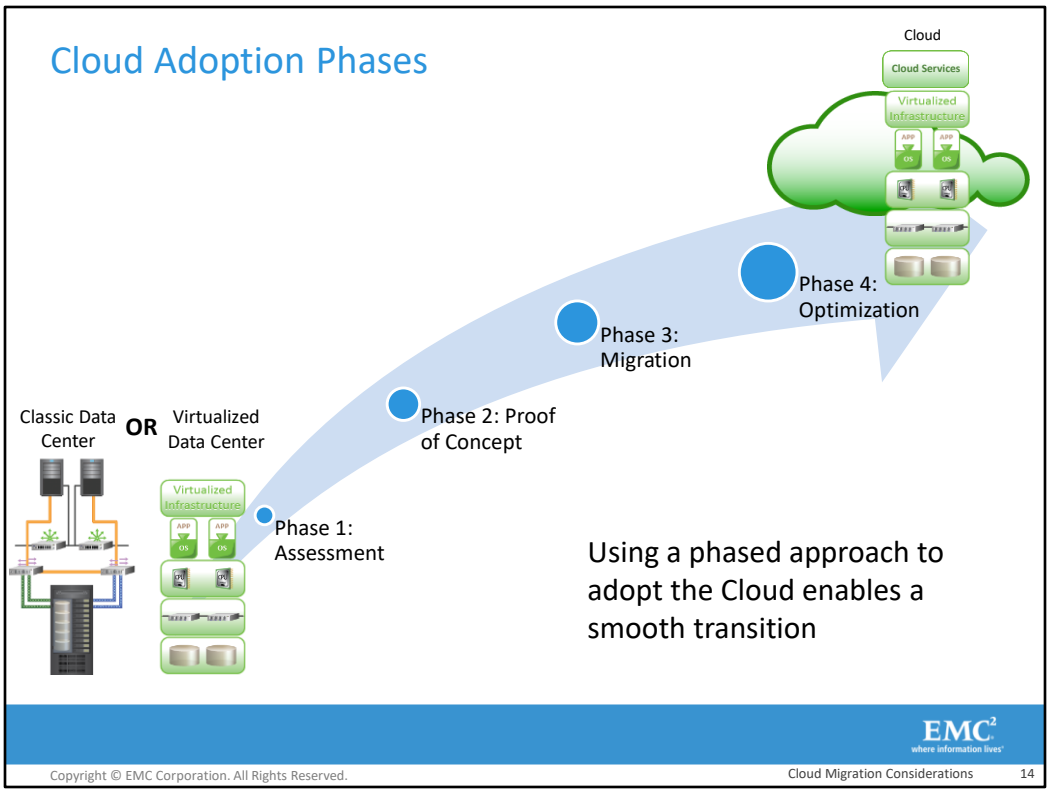
When using open standards, choose technologies that are widely accepted and proven. For example, Use of common APIs for an application may allow an organization to move it to other Cloud with minimal or no change. Similarly, use of Open Virtual Machine Format (OVF) is a common VM format that enables using a VM built in one Cloud to be deployed to another Cloud with minimum or no changes.

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This lesson covers the different phases involved in adopting the Cloud. The adoption phases are: assessment phase, proof of concept phase, migration phase, and optimization phase.

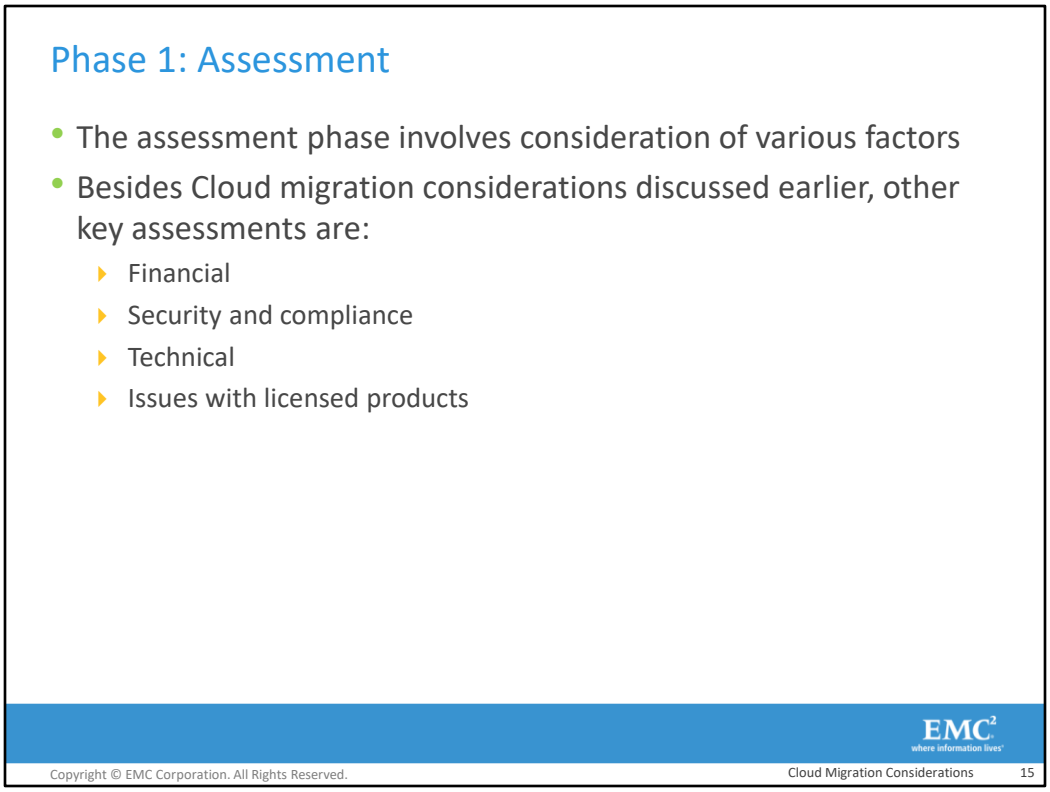
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After identifying the right application for the Cloud, it may be moved to the Cloud. Organizations typically adopt the Cloud in phases for smooth transition. The adoption process typically consists of four phases. They are assessment phase, proof of concept phase, migration phase, and optimization phase.

Each phase is discussed in detail in the next few slides.

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The first phase in Cloud adoption is the assessment phase. To ensure successful assessment, it is important to define and understand its objectives. Assessment involves consideration of various factors. Assessment should be performed for each application that is identified as a potential candidate for the Cloud. Besides considerations discussed in the previous lesson – Migration Considerations, other key assessments are: financial assessment, security and compliance assessment, technical assessment, and assessment of issues related to migration of licensed products.

Each of these assessments is discussed in detail in the next few slides.

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Financial Assessment

- Provides cost comparison of in-house vs. service provider
  - TCO and ROI
- Requires cost consideration of the following elements:

Saving		Spending
CAPEX	OPEX	Other Cost
• Servers • Storage • Operating System (OS) • Application • Network equipments • Real estate	• Power and cooling • Personnel • Bandwidth • Maintenance • Support • Backup	• Migration • Compliance and Governance • Service cost

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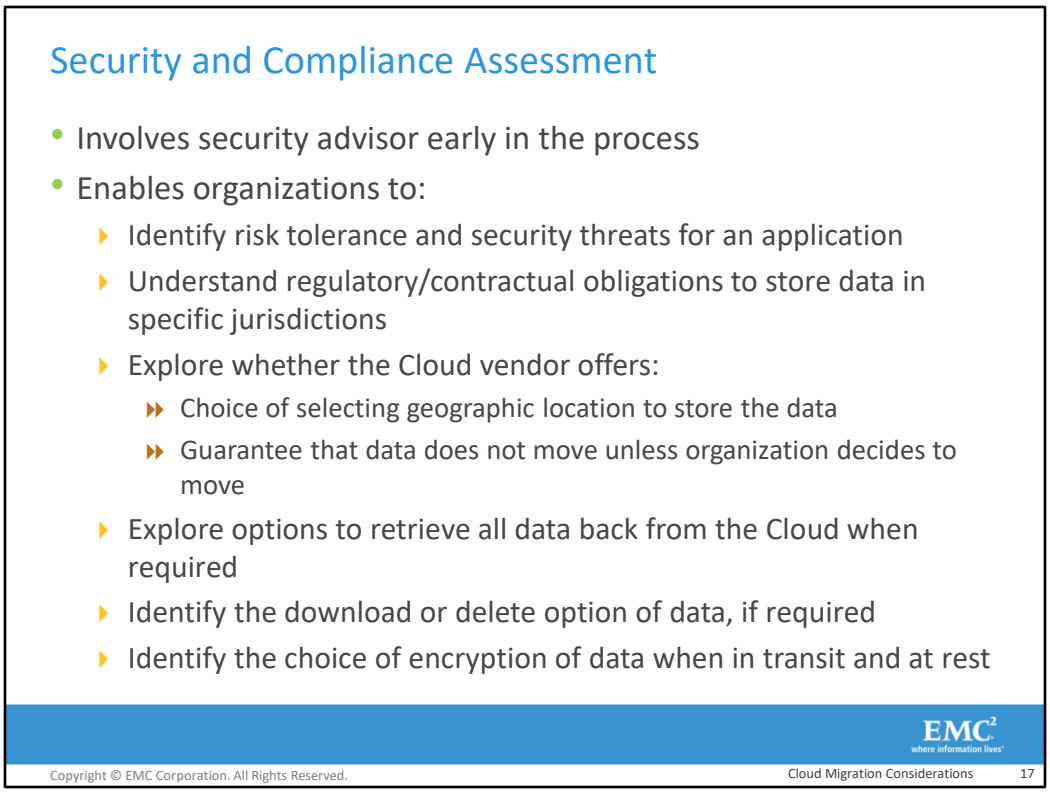
Financial assessment helps in determining Total Cost of Ownership (TCO) and Return on Investment (ROI) and in further building a business case. Financial assessment requires detailed and careful analysis, because it provides a clear picture of the cost involved in owning and operating a data center versus employing a Cloud based infrastructure.

Financial assessment is not simple and requires considering Capex, Opex, and Overhead cost.

For example, consider an organization that requires moving an application that has specific requirements, such as archiving data and retaining it for long period in the Cloud (compliance and governance) or specific security requirements. Then, the following are the cost components that should be considered to perform a financial assessment:

1. CAPEX includes cost of servers, storage, Operating System (OS), application, network equipments, real estate, etc. Moving an application to Cloud reduces CAPEX, compared to maintaining it on site.
2. OPEX includes the cost incurred for power and cooling, personnel, bandwidth, maintenance, support, backup, etc. Moving an application to Cloud reduces OPEX, compared to maintaining it on site.
3. Overhead includes migration cost and cost to ensure compliance and governance. In this case, the organization may have to pay an additional cost for archival service and storage capacity (to ensure compliance and governance). The organization may have to pay an additional cost to get the required security services for their application. Moving an application to Cloud may incur overhead cost over and above the subscription fees.

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Data security can be a daunting issue if not properly understood and analyzed. Hence, it is important to understand risks and threats. Based on the sensitivity of data, classify the data assets into different categories (confidential, public, internal only, etc). This will help identify which data asset can be moved to the Cloud and which can be kept in-house.

If an organization has strict security policies and compliance requirements, it is recommend to involve the organization's security advisers and auditors early in the process.

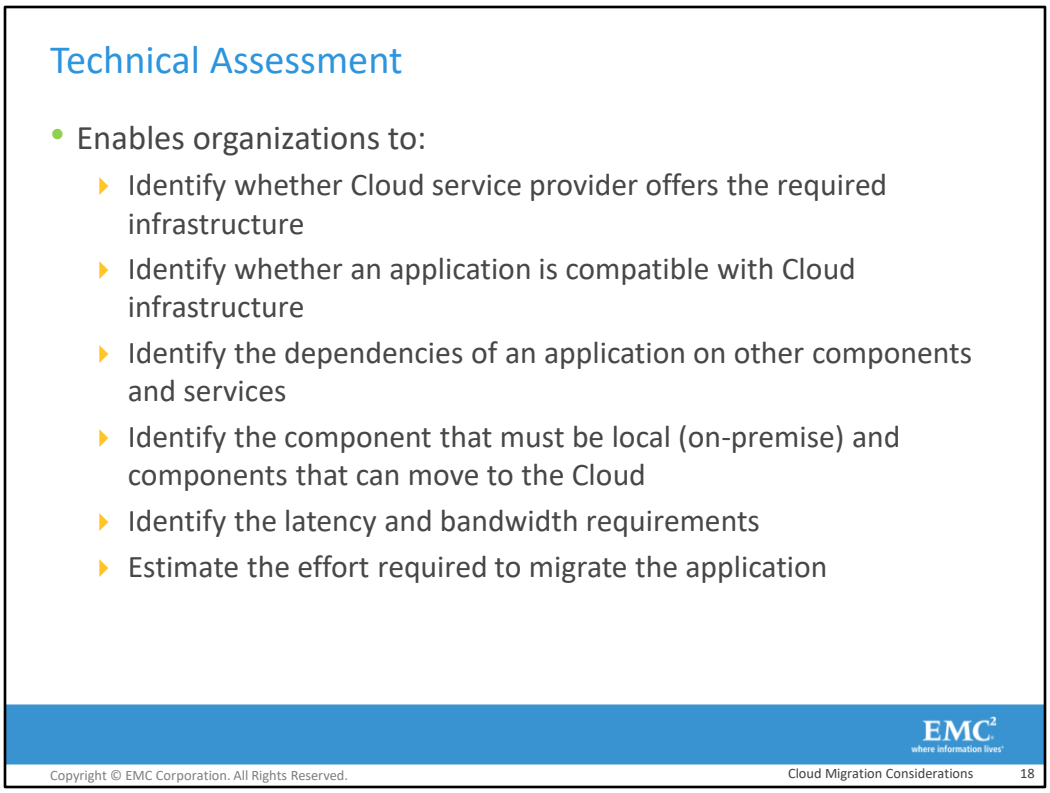
This assessment will enable organizations to:

- Identify the overall risk tolerance for an application.
- Identify the security threats that have a likelihood of materializing into actual attacks.
- Understand the regulatory or contractual obligations to store data in specific jurisdictions.
- Explore whether the Cloud vendor offers a choice of selecting the geographic location to store the data and a guarantee that the data does not move unless the organization decides to move it.
- Explore options if the organization decides to retrieve all of the data back from the Cloud.
- Identify whether the Cloud vendor offers options to download or delete the data whenever required.
- Identify the choices offered by the Cloud vendor to encrypt the data while it is in

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transit and while it is at rest.

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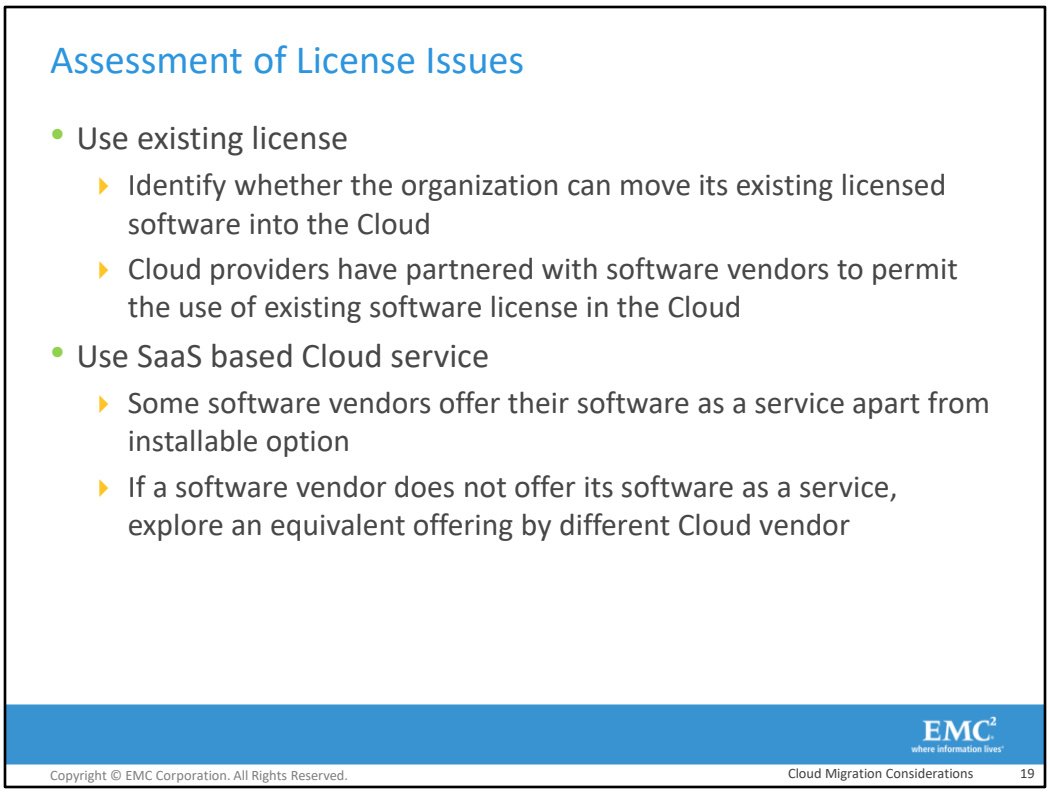
A technical assessment helps identify the applications that are more suited to the Cloud. It also helps the organization determine the applications that should move into the Cloud first, those that should move later, and those that should remain in-house.

Technical assessment identifies the dependencies of an application on other components and services. For example, a Web-based application depends on the database, login, and authentication services running on another system. In most cases, the best candidates for the Cloud are the applications that have minimum dependencies.

This assessment will enable organizations to:

- Identify whether the Cloud service provider offers all of the required infrastructure building blocks.
- Identify whether the application can be packaged into a Virtual Machine (VM) instance and be run on a Cloud infrastructure.
- Identify the component that must be local (on-premise) and components that can move to the Cloud.
- Identify the latency and bandwidth requirements.
- Estimate the effort required to migrate the application.

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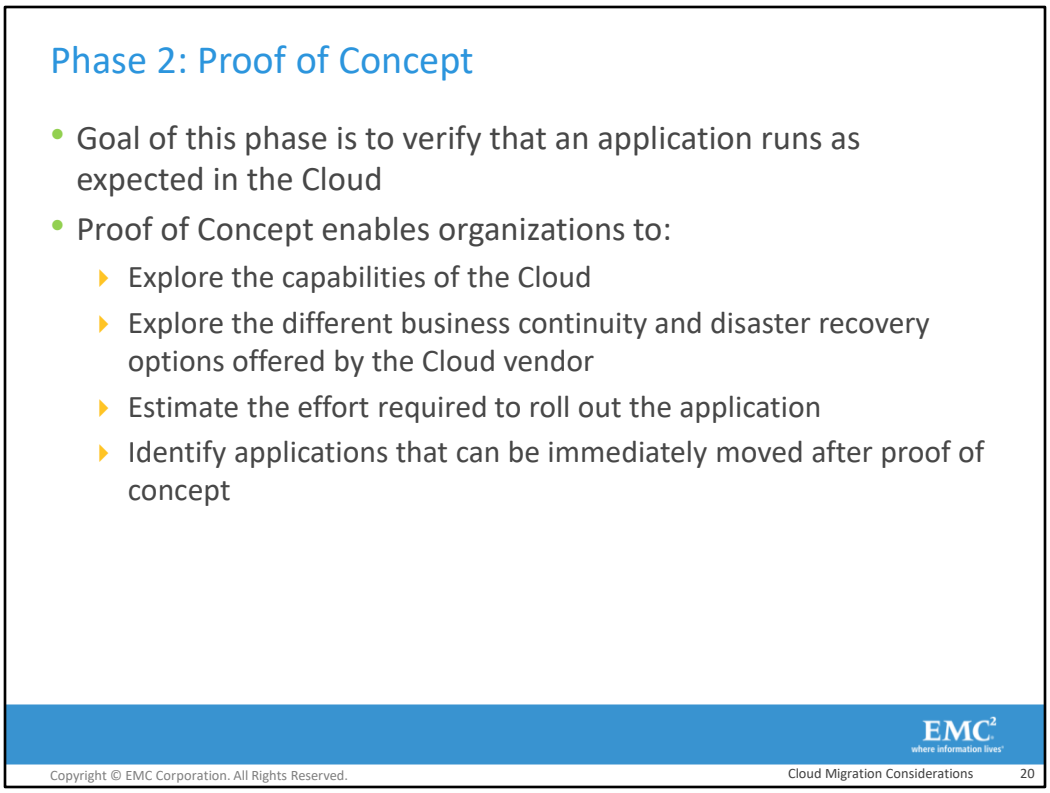


It is important to assess the issues related to migrating licensed software to the Cloud at an early stage.

There are two options available for migrating licensed software to the Cloud:

- **Use the existing license:** Cloud providers have partnered with several software vendors. Due to this partnership, software vendors permit organizations to use their existing product license on Cloud. This option offers the easiest path to move licensed software to the Cloud. In this option, the organization purchases license in the traditional way or uses the existing license in the Cloud.
- **Use SaaS based Cloud service:** Many software vendors offer their software with two options, one that can be installed on-premise, and the other as a service. In this option (SaaS), the existing on-premise installed application is migrated to a hosted offering (SaaS) by the same vendor. If a software vendor does not offer its software as a service, explore and migrate the data on an equivalent offering by a different Cloud provider. In this case, the Cloud provider may charge a monthly subscription fee.

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After a thorough assessment, identifying the right candidate for the Cloud, and estimating the efforts required for migration, it is time to test the application with a proof of concept. This phase helps to understand what an application can do and cannot do in Cloud.

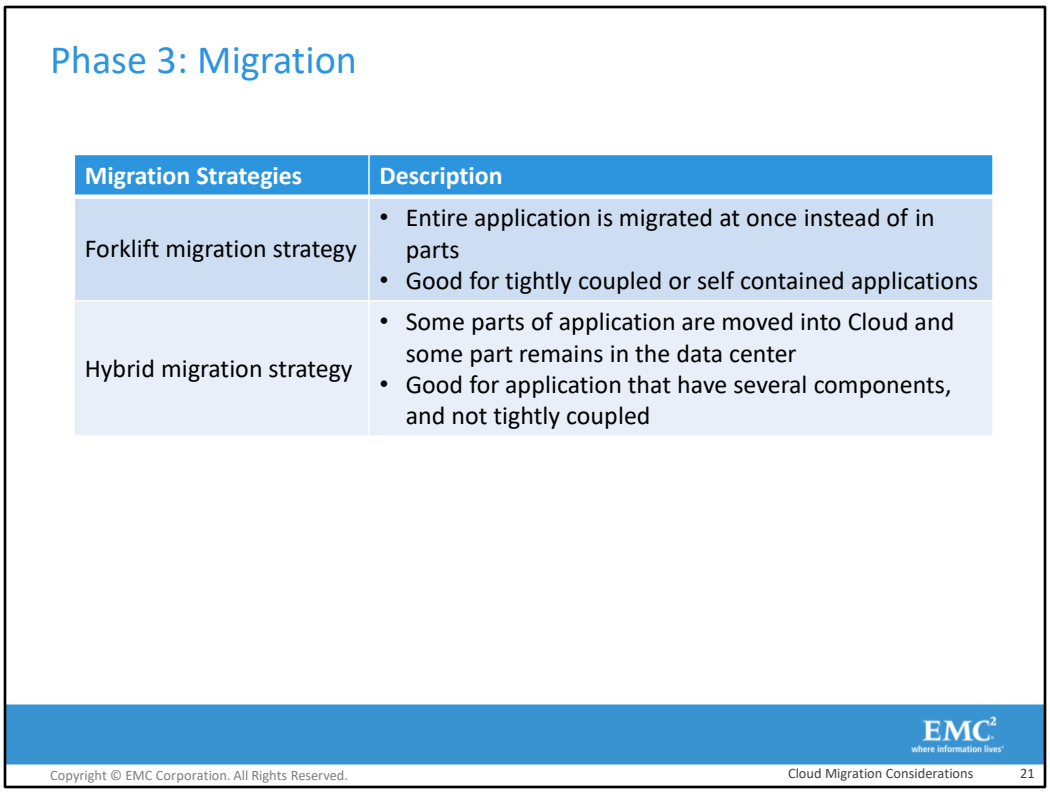
The goal of this phase is to check whether an application runs as expected after migrating it to the Cloud. It is recommended to perform a thorough testing of the application during this phase. In this phase, the organization can validate the Cloud technology, test legacy software in the Cloud, perform necessary benchmarks, and set expectations.

This assessment will enable organizations to:

- Explore the capabilities of the Cloud
- Explore the different business continuity and disaster recovery options offered by the Cloud vendor
- Estimate the effort required to roll this proof-of-concept out to production
- Identify applications that can move after proof of concept

After this phase, the organization will gain hands-on experience with the Cloud environment, which in turn gives them more insight into what hurdles need to be overcome in order to move ahead.

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In this phase, applications are migrated to the Cloud. There are two application migration strategies:

- **Forklift Migration Strategy:** In this strategy, rather than moving applications in parts over time, all applications are picked up at once and moved to the Cloud. Tightly coupled applications (multiple applications that are dependent on each other and cannot be separated) or self-contained applications might be better served by using the forklift approach. Self-contained Web applications that can be treated as a single entity and backup/archival systems are examples of systems that can be moved into the Cloud using this strategy.
- **Hybrid Migration Strategy:** In this strategy, some parts of the application are moved to the Cloud while leaving the other parts of the application in place. The hybrid migration strategy can be a low-risk approach to migration of applications to the Cloud. Rather than moving the entire application at once, parts of it can be moved and optimized, one at a time. This strategy is good for large systems that involve several applications and those that are not tightly coupled.

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Phase 4: Optimization

- Test the application after migration is complete
- Understand the usage pattern and optimize resource consumption
- Relinquish idle resources

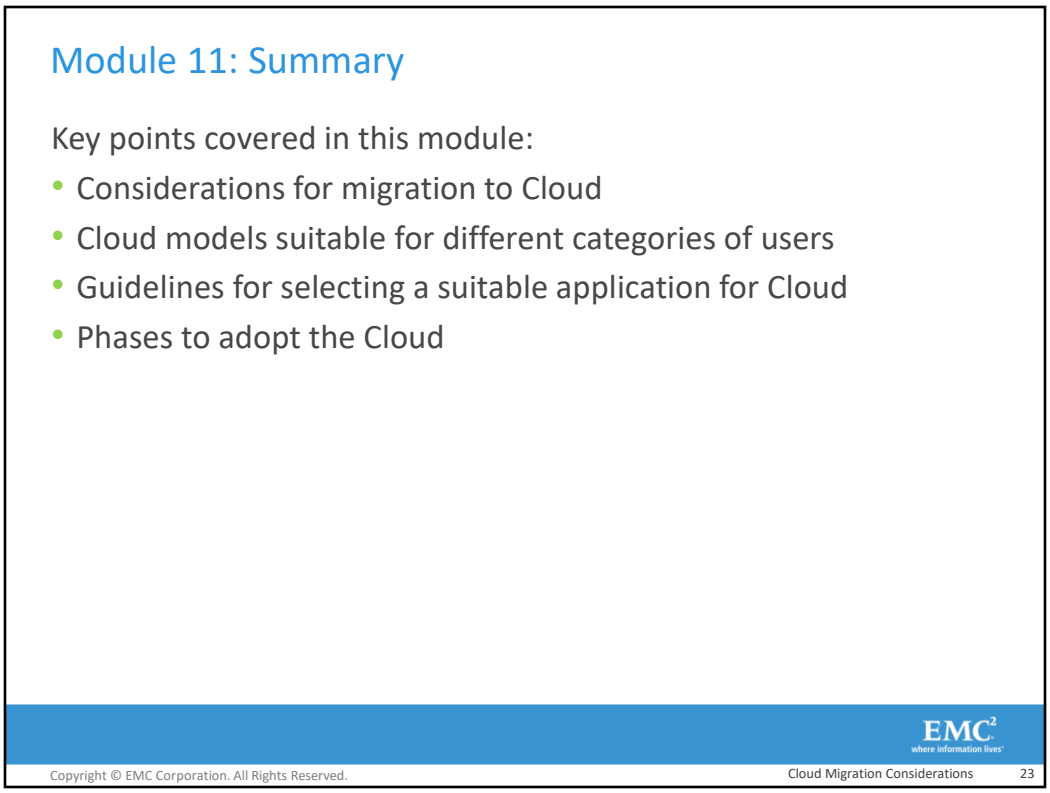
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After migrating the application to the Cloud, run the necessary tests and confirm that everything is working, as expected. In this phase, focus on how to optimize the Cloud based application in order to increase cost savings.

Understand the usage pattern to optimize the resources consumed. To understand the usage pattern, monitor the resources consumed and the workload. Based on the workload, resources can be scaled up or scaled down. For example, if a customer-facing Website, deployed on a Cloud infrastructure, does not expect any traffic from certain parts of the world at a certain time of the day, the resources consumed by that region may be scaled down for that time.

Inspect the system logs periodically to understand the usage of the resources. Relinquish the idle resource.

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This module covered the migration of an application to the Cloud.

The key points to consider while migrating to Cloud are Criteria for Cloud vendor selection, Service Level Agreements (SLAs), Cloud performance consideration, Cloud vendor lock-in, and Cloud open standards.

There are three Cloud models such as Public, Private, and Hybrid. Based on the business type, a specific model is selected. Individuals and startups typically prefer Public Cloud, SMB prefers Hybrid Cloud, and enterprises prefer Private Cloud.

The guidelines for selecting a suitable application for Cloud are:

- Proprietary and mission-critical application: Should remain in-house
- Non-proprietary but mission-critical application: Can be moved to Cloud only if the organization lacks the skills to maintain the application or if the maintenance cost is high
- Non-proprietary and non-mission critical applications: Is a good candidate for Cloud

The Cloud adoption process typically consists of four phases, such as the assessment, proof of concept, migration, and optimization phases.

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Check Your Knowledge

1. What are the key concerns that organizations have while adopting the Cloud?
2. Describe the guidelines for choosing the right application for the Cloud.
3. What are the two options available for migrating licensed software to the Cloud?
4. Describe the two migration strategies.
5. What should be considered to avoid Cloud vendor lock-in?

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